

Forget robots - programmable cells are coming



Edward Henning, Editor-in-Chief

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You may have caught recent news stories about some rather over-the-top statements at a conference to the effect that robots would in the future pose serious risks to humanity. There seems to have developed something of a pattern here, with some scientists - if they still deserve the title - drumming up fear, presumably to attract more grants for research. You can probably tell that I don't think these stories need be taken too seriously.

Another contender often cited as representing a threat to humanity is the grey goo that could result from self-replicating nano-bots getting out of hand and smothering us all. This all seems to assume that the creators of these devices will have the level of intelligence of a motor manufacturer that builds a car with a powerful accelerator, and completely forgets to install a brake. Not very likely.

But there is one trend in technology that I think could well pose dangers - this lies with what we might call digital molecular biology.

An old colleague once summed up the progress of modern technology quite neatly: "Once you can digitise something, you then have control over it." I have often found this a useful way of describing the progress of computing. First, we managed to gain control over text, first in word processors, and a little later, databases. Simple numbers were next, with the rise of spreadsheets and the like. Next came simple graphics, sound, and more recently video. We have not yet fully mastered video - it is still often a very expensive procedure in terms of processing power, but we are well on the way to being able to edit and manipulate video as easily as we now can text and audio.

Next will be 3D and virtual worlds, and it appears, biology. There are two sides to this.

One is the development of biomolecular computers - computing devices made of DNA. The most impressive feat that I have read of for one of these devices was this: it was able to detect the presence of cancer in a test tube and dispatch suitable molecules to kill it. Imagine that mechanism packed in a virus-like shell and injected into the bloodstream of a patient. We are certainly moving in that direction, and quickly.

The other side of this development concerns what we might call the reading and writing of DNA information. Remember the huge fuss - and cost - just a few years ago when the human genome was first mapped?

One scientist recently stated that until a few years ago, improvements in gene sequencing had developed at a similar pace to silicon technology - rather like Moore's Law. However, recently, things have speeded up very dramatically - improvements multiplying by a factor of ten every year. Just a few years ago, the cost of (almost completely) reading the human genome was around \$3 billion. Today it is \$5,000, and falling fast.

So, we are not far away from cheaply being able to read entire genomes into a computer, manipulate them, and then write them back out into new DNA. Where might we put that DNA? Certainly into the virus package that should go and find cancer cells and kill them, and leave everything else alone. Enhancing a human egg will be the next step, and then somebody will also want to put it into an Ebola or HIV virus so that it will only target specific racial groups and leave everybody else unharmed. Any powerful technology can be used for both good and evil, from simple axes and knives to nuclear power and digital biology. But, I am more concerned about the possible mistakes that might arise from this technology.

I have long been opposed to genetic modification, mainly because it is such a hit and miss affair. New genes are crudely shoved inside a cell nucleus, and another gene put in there to start it up. The methods are very much blunt instruments. A living cell is such an incredibly complex system, as is the biosphere into which, say, a GM version of cotton might be introduced, that we cannot possibly predict all the effects that might follow. At least, we cannot be certain that ill effects will not occur.

Life is tenacious and powerful, and bad effects can cause enormous problems and suffering. If we introduce something into the biosphere that turns out to cause harm, it might prove impossible to contain. If we are approaching the stage where we can directly and easily manipulate the very coding, the DNA, that controls life, then the possibilities are enormous. The benefits could clearly be considerable, but are they worth the risks that would accompany them? Accelerating our own evolution might seem exciting, but how will we handle the risks of accelerated extinction?

A handwritten signature in black ink that reads "E. Henning".

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